

Broker License Benchmark Experimental Protocol

Goal

- Build a benchmark for model performance on California P&C broker-agent licensing knowledge and applied broker reasoning.
- Start with multiple-choice licensing questions, then expand to expert-style reasoning and broker workflow tasks.
- Measure base LLMs, agent harnesses, retrieval setups, and post-training variants.

Non-Goals For This Project

- Life & Health broker tests.
- Commercial Lines broker tests.
- State-by-state licensing beyond the minimum needed to compare California against other P&C exams.
- Production legal/compliance advice.

Benchmark Scope

- Domain: California Property & Casualty broker-agent licensing.
- Core task: answer licensing-style multiple-choice questions.
- Applied task: reason like an expert broker on scenarios, compliance constraints, disclosures, and distribution strategy.
- Future scope: other states, non-resident licenses, surplus lines.

Example benchmarks:

- [arXiv benchmark paper](#)
- [InfoQ: evaluating AI agents lessons learned](#)
- [Mercor APEX](#)
- [Mercor APEX-Agents](#)
- [OpenAI GDPval](#)
- [OpenAI Evals](#)
- [Vals AI Benchmarks](#)

Data Sources

- Official California Department of Insurance exam objectives and licensing materials.

- Official P&C exam objectives from a small set of comparison states.
- Online P&C practice exams and mock tests.
- Prep provider question banks, if licensed or otherwise permitted.
- Public educational content on P&C concepts, state insurance code, ethics, and broker-agent obligations.
- Expert-written questions from licensed brokers / compliance experts.
 - We can hire licensed brokers on Fiverr/Upwork.
- Our client sales transcripts, if shared and permissioned.

Data Acquisition

- Crawl or scrape mock test content where terms permit.
- Store question, answer choices, correct answer, explanation, source, topic, difficulty, and license line.
- Track provenance for every question.
- Deduplicate near-identical questions.
- Split into train/dev/test before augmentation.
- Keep a clean held-out test set that is never used for SFT, RL, prompt tuning, or retrieval tuning.

Question Types

- Multiple-choice exam questions:
 - Single best answer.
 - Ground-truth answer key.
 - Optional explanation.
- Scenario variants:
 - Same concept, different insured, carrier, risk, disclosure, jurisdiction, or timeline.
 - Tests robustness beyond memorization.
- Rubric reasoning tasks:
 - Free-form answer.
 - Must identify applicable rule.

- Must apply the rule to facts.
- Must avoid unsafe or non-compliant recommendation.
- Must state uncertainty or escalation path when appropriate.
- Sales transcript tasks:
 - Identify what can be asked.
 - Identify what should not be represented.
 - Recommend next broker action.
 - Flag licensing/compliance risks.

Rubric Design

- Score MCQ with exact-match accuracy.
- Score reasoning with expert rubrics:
 - Correct legal/insurance concept.
 - Correct application to facts.
 - No hallucinated authority.
 - Appropriate compliance caution.
 - Practical broker-quality recommendation.
- Use binary criteria where possible, inspired by agent benchmarks.
- Use expert adjudication for a gold subset.
- Use model-as-judge only after calibration against expert labels.

Data Augmentation

- Generate scenario mutations while preserving the same underlying rule.
- Vary:
 - Personal vs commercial insured.
 - Property vs casualty line.
 - Agent vs broker authority.
 - Carrier-appointed vs broker-bond context.
 - Resident vs non-resident licensing.
 - California vs other-state distribution.
- Validate augmented questions with an expert or high-confidence rule check.
- Do not let augmented examples leak into held-out eval if generated from test seeds.

Cross-State Discrepancy Analysis

- Compare California P&C exam objectives against 5-10 other states.
- Look for differences in:
 - License naming: producer, agent, broker-agent, broker.
 - Property and Casualty as combined vs separate exams.

- State-specific insurance code and ethics requirements.
- Pre-licensing education hours.
- Passing score and exam length.
- Fingerprinting / background checks.
- Broker bond or appointment requirements.
- Non-resident reciprocity rules.
- Surplus lines treatment.
- Personal lines vs broader P&C coverage.
- Create labels:
 - `universal_p_c_concept`
 - `california_specific_rule`
 - `state_specific_rule`
 - `reciprocity_rule`
 - `potential_conflict_between_states`
- Use this to test whether models overgeneralize from national P&C knowledge into California-specific answers.

Eval Pipeline

- Dataset format:
 - `id`
 - `source`
 - `question`
 - `choices`
 - `answer`
 - `explanation`
 - `topic`
 - `difficulty`
 - `license_line`
 - `split`
 - `rubric`
- Runner:
 - Standard prompt.
 - Chain-of-thought hidden or disabled depending on provider.
 - Optional retrieval context.
 - Temperature 0 for exam accuracy.
 - Multiple seeds for agentic or stochastic systems.
- Metrics:
 - Accuracy.
 - Topic-level accuracy.
 - Calibration / confidence.
 - Explanation quality.
 - Compliance safety.

- Cost and latency.
- Pass/fail estimate versus exam threshold.

Model And Agent Matrix

- Frontier LLMs:
 - OpenAI models.
 - Anthropic models.
 - Google Gemini models.
 - xAI / other frontier models.
- Open-source LLMs:
 - Qwen.
 - Llama.
 - DeepSeek.
 - Mistral.
 - Kimi / Moonshot.
- Harness variants:
 - Base model, no tools.
 - RAG over official licensing docs.
 - RAG over licensed prep corpus.
 - Agent with search + calculator + citation requirement.
 - Self-consistency / majority vote.
 - Critic or verifier model.
- Training variants:
 - Base model.
 - SFT on training questions.
 - SFT on explanations.
 - RL on MCQ correctness.
 - RL or DPO on expert rubric preferences.
 - Domain-adapted retrieval without weight updates.

Experiments

- Baseline zero-shot MCQ accuracy.
- Few-shot exam prompt.
- RAG with official CDI materials only.
- RAG with prep corpus.
- SFT vs RAG vs SFT+RAG.
- RL after SFT.
- Expert reasoning rubric eval before and after post-training.
- Robustness on augmented scenario questions.
- Cross-state transfer: California-trained model on other-state licensing questions.
- Cross-state conflict test: same scenario, different state, different correct answer.

- Sales transcript recommendation quality.

Risks And Controls

- Copyright / terms risk from scraped practice tests.
- Contamination risk if public questions are already in model pretraining.
- Shortcut learning from answer-choice artifacts.
- Model-as-judge bias.
- Legal/compliance sensitivity: benchmark should not present outputs as legal advice.
- Keep clean provenance, permissions, and held-out expert-created questions.

Immediate Next Steps

- Collect official CDI exam objectives and PSI candidate handbook links.
- Pick comparison states for discrepancy analysis.
- Collect official P&C exam outlines for comparison states.
- Inventory 10-20 online mock exam sources and their usage rights.
- Draft the canonical dataset schema.
- Create 25 expert-written seed questions across core P&C topics.
- Create 10 reasoning rubrics for broker-like scenarios.
- Decide first model sweep: frontier models + Qwen + Llama.